

Constructive Method on the Goldbach Conjecture, Proving that the Cramer-Granville Conjecture Implies Goldbach's for Practically all Large Evens

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Abstract

In the present paper we humbly try to formalize a constructive method that should generate an infinite amount of Goldbach pairs (p, q) such that $p + q = 2n$ for some n . We will also use some established theorems and conjectures on the distribution of prime numbers, particularly Hoheisel's theorem on prime gaps and its refinements. By constructing a systematic method to derive new Goldbach pairs from existing ones, we establish a lower bound on the density of even integers satisfying the Goldbach conjecture. In the end we will try to derive a minimum guaranteed lower bound on the number of valid Goldbach representations up to a given even integer. Ultimately, we demonstrate that under the Cramer-Granville refined conjecture on prime gaps, the strong Goldbach conjecture holds for practically all sufficiently large even integers.

1 Construction Method

Let $(n, p, q, p', q', c) \in (\mathbb{N}^*)^6$

With p and q prime such that $p + q = 2n$

With p' and q' prime such that $p' + q' = 2n + 2c$

Let $k \in \mathbb{Z}$

We pose $p' = p + 2k$ and $q' = q - 2k + 2c$

2 Utilizing Established Prime Distribution Theorems

We know from Hoheisel's theorem:

There exists $\theta \in (1/2; 1)$ such that for any large enough $m \in \mathbb{N}$, there always exists a prime number $m' \in (m; m + m^\theta]$ and $m'' \in (m - m^\theta; m]$

Know that the best unconditional proved value of θ is $\theta = 0.565$ and best conditional (under R.H) is $\theta = 0.5 + \varepsilon$

From this theorem we have the existence of p' prime such as $p' \in (2n - 5 - (2n - 5)^\theta; 2n - 5]$

From which $p + q - 5 - (2n - 5)^\theta < p + 2k \leq p + q - 5$

So $\frac{q-5-(2n-5)^\theta}{2} < k \leq \frac{q-5}{2}$

Lemma 1. $p' = p + 2k$ can be prime with $k \in \left(\frac{q-3-(2n-5)^\theta}{2}; \frac{q-5}{2} \right]$

So now we have the guaranteed existence of p' prime that only depends on k that itself can exist

On top of this we have the existence of q' prime such as $q' \in (q - 2k; q - 2k + (q - 2k)^\theta]$

From which $q - 2k < q - 2k + 2c \leq q - 2k + (q - 2k)^\theta$

So $0 < c \leq \frac{(q-2k)^\theta}{2}$

Lemma 2. $q' = q - 2k + 2c$ can be prime with $c \in \left(0; \frac{(q-2k)^\theta}{2} \right]$

So now we have the guaranteed existence of q' prime that only depends on c that itself can exist depending on k so as long as $\frac{(q-2k)^\theta}{2} \geq 1$ which is always true

Let's study the worst-case scenario for c

We have $c_{\max} = \frac{(q-2k_{\min})^\theta}{2} = \frac{(q-q+5+(2n-5)^\theta)^\theta}{2} = \frac{(5+(2n-5)^\theta)^\theta}{2}$

And as n grows $c_{\max} \approx \frac{(2n)^{\theta^2}}{2} = n^{\theta^2} \cdot 2^{\theta^2-1}$

Lemma 3. There exists a pair of primes (p', q') originally constructed from a pair of primes (p, q) such that $p + q = 2n$ and $p' + q' = 2n + 2c$ with $c \in (0; n^{\theta^2} \cdot 2^{\theta^2-1}]$

3 Calculating the Guaranteed Lower Bound

Let a be the counter of successive Goldbach evens guaranteed under our method

We know that $\Delta_{\max} = 2c_{\max} = n^{\theta^2} \cdot 2^{\theta^2}$

So $2n_a + 1 \leq 2n_a + \Delta_{\max}$

Which means $2n_a + 1 \leq 2n_a + (2n)^{\theta^2}$

So in the worst case $2n_a + 1 = 2n_a + (2n)^{\theta^2}$

Let's treat n as a continuous function of a , we admit that this leap needs rigorous proof, but for the moment we are sorry to skip this step. However we need to admit that the function is pseudo-arithmetic sequence when it's in the worst case (which is exactly what we are studying) almost certainly guarantees us our leap.

We will then have $\frac{dn}{da} = (2n)^{\theta^2}$

So $n^{-\theta^2} \cdot dn = 2^{\theta^2} \cdot da$

Let's integrate both sides:

$$\int n^{-\theta^2} dn = \frac{n^{1-\theta^2}}{1-\theta^2} + c$$

$$\int 2^{\theta^2} da = 2^{\theta^2} \cdot a + c'$$

For simplicity let's put $c = c'$

We then get:

$$\frac{n^{1-\theta^2}}{1-\theta^2} = 2^{\theta^2} \cdot a$$

So $a = \frac{n^{1-\theta^2}}{(1-\theta^2) \cdot 2^{\theta^2}} = O(n^{1-\theta^2})$

Theorem 1. Until a certain $2n$, we have the least guaranteed amount a of valid Goldbach evens:

$$a \geq O(n^{1-\theta^2}) \text{ with } C = \frac{2^{-\theta^2}}{1-\theta^2}$$

4 Applications

Under the best unconditional proven result, we have:

$$a \geq 1.14 \cdot n^{0.724375}$$

Under the assumption of R.H we have:

$$a \geq O(n^{0.75})$$

Under the assumption of Cramer-Granville's refined conjecture:

First we need to "translate" the conjecture's statement to the form of $(x; x + x^\theta]$

As n grows to infinity, the gap between the n -th and $n + 1$ -th prime is equal or less than $O(\ln^2 p_n)$ with p_n the n -th prime let $x = p_n$

We solve: $x^\theta = O(\ln^2 x)$

We get: $x^\theta = C' \cdot \ln^2 x$

Then: $\theta = \log_x(C' \cdot \ln^2 x)$

So: $\theta = \log_x C' + 2 \log_x \ln^2 x = \frac{\ln C' + 2 \ln \ln x}{\ln x}$

So: $\lim_{x \rightarrow +\infty} \theta = 0$

Which means that: $k \geq O(n)$ which is the essence of Goldbach's conjecture $C = 1$ here.

Theorem 2. *Goldbach's conjecture holds under Cramer-Granville's refined conjecture for practically every large enough even $2n$.*

5 Questions

Can we let $\theta \in \mathbb{C}$?

6 References

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